Arunabha Majumder



Mechanical | Mechatronics Engineer

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PROFILE

Mechanical engineer with a strong base in modelling, calculations and test validation. In my PhD I worked on stiffness modelling and Jacobian-based force estimation for a parallel robot, checked against force sensor measurements. I designed the load-carrying parts of a shoulder exoskeleton, optimised the design with a genetic algorithm and measured its stiffness on a physical test setup. I use MATLAB daily for calculations and data processing, and I write clear technical reports and papers. PhD in Mechatronics and Robotics at Aalborg University (thesis expected March).

CORE SKILLS

- **Modelling and calculation:** Stiffness modelling, kinematics and Jacobian-based force estimation, validated against sensor data
- **Analysis and optimisation:** Stiffness analysis in SolidWorks; design optimisation with a genetic algorithm
- **Scripting and tools:** MATLAB/Simulink (daily use), Python (basic), LaTeX
- **Testing and validation:** Test rigs, force/torque sensors, Qualisys motion capture and data acquisition, test data processing
- **CAD:** SolidWorks (CSWA): parts, assemblies, technical drawings; design of load-carrying parts
- **Reporting:** Technical reports, conference papers and presentations; sketches and visuals for papers and demos

PROFESSIONAL EXPERIENCE

Research Engineer, Shoulder Exoskeleton (PhD project VIEXO) | 2023 - Present
Aalborg University, Aalborg, Denmark

- Came up with and designed a hybrid shoulder exoskeleton where a spring carries most of the arm load and a small 6 Nm motor adds the rest; it cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Owned the mechanical design from hand sketches to SolidWorks models and drawings, then 3D printed and rebuilt parts many times to improve fit, strength and ease of assembly.
- Optimised the exoskeleton design with a genetic algorithm and measured the stiffness it provides on a physical test setup.
- Integrated the motor, encoder and force sensors into the frame and worked with control and software colleagues to get it running as one system; built custom test rigs.
- Collected data from 12 participants (EMG with the Qualisys data acquisition system and motion capture).
- Worked in a team of 3 on an Assist-as-Needed controller (my part: equations such as the Fourier fit, and defining how the GUI should behave); muscle effort dropped by more than 15%.
- Estimated tip force on a 5-bar parallel robot from its variable-stiffness actuators (within 10% RMS error vs a force sensor), so no extra sensor was needed; implemented an Arduino-based trigger to sync robot motion with Qualisys cameras for arm impedance tests (6 participants).
- Taught university courses and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Pneumatic Artificial Muscles (Master's thesis) | 2020 - 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated and assembled pneumatic muscles to an existing lab design, collected test data on the lab's setup and used it to design the outer PID loop of a gain-scheduled controller (0.3-0.78% position error, IEEE CONECCT 2022).

Bachelor's Capstone, Upper-Limb Rehabilitation Exoskeleton | 2018 - 2019
Siddaganga Institute of Technology, Tumakuru, India

- Led CAD design, machining, assembly and testing; Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant.
- Student teams: built a Go-Kart (2017) and an Efficycle hybrid vehicle (2018).

EDUCATION

PhD, Mechatronics and Robotics (defence expected in 2027) | 2023 - Present
Aalborg University, Department of Materials and Production, Denmark

Study abroad, National University of Singapore (NUS), Sep - Nov 2025

MTech, Mechatronics | 2020 - 2022
AcSIR - CSIR-CMERI, Durgapur, India

BTech, Mechanical Engineering | 2015 - 2019
Siddaganga Institute of Technology (VTU), India

Diploma in Fine Arts (Painting)
Bangiya Sangeet Parishad, India

AWARDS AND PUBLICATIONS

- Best Research Paper Award, 9th IFToMM ISRM (2026); Certified SolidWorks Associate (CSWA); Best Major Project Award and TATA Technologies Innovation Grant (2019)
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer. Also IEEE CONECCT 2022.

LANGUAGES

English (C1), Bengali (native), Hindi (spoken)

REFERENCES

Available upon request.